

CHUKWUEMEKA OSMUND AZODO

MACHINERY ENGINEER | ROTATING EQUIPMENT, CONTROLS & RELIABILITY | OIL & GAS

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B.Eng Mechanical Engineering | ASQ Certified Reliability Engineer (CRE) | COREN R.Engr | ASME, SPE, NSE, ASQ

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Recruitment Team
ExxonMobil Nigeria
Esso Exploration and Production Nigeria Limited, Lagos

Application: Machinery Engineer, Nigeria Deepwater Assets

Dear Hiring Team,

I am applying for the Machinery Engineer position supporting the Nigeria Deepwater assets. I am a mechanical engineer with over ten years in machinery and rotating equipment engineering, currently the Machinery Engineer for the Oso Gas Hub, the largest offshore gas processing facility in the Seplat joint venture, and part of the former Mobil Producing Nigeria asset base. The standards, work processes and much of the machinery I work to every day are ExxonMobil-legacy, so I would join already fluent in how your assets are engineered and governed.

Safeguarding machinery reliability and availability is the substance of my current role. I review equipment performance daily: vibration spectra and trends, lube-oil analysis, compressor performance maps, turbine parameters and pump curves across HP and LP centrifugal compressors, turbo-expanders, gas turbines, cryogenic and API 610 pumps, via SolarInsight, Seeq and PI XHQ, and I issue the analysis and the recommendation to operations and maintenance whenever a machine drifts from its expected envelope.

I act as technical lead for machinery troubleshooting and recovery. When a bundle change-out left a 130 MMSCFD compressor with severe vibration, I led the root cause analysis with the OEM and in-house teams, mapped the contamination paths and ran sectioned borescope inspection of the cooler bundles; the cause proved to be debris-induced rotor unbalance, and piping cleaning, lube-oil flushing and re-inspection returned the machine to service within ISO vibration limits. A separate fleet-wide RCFA (fifteen years of records, using pooled failure-rate and Crow-AMSAA reliability-growth analysis) let me show the affected compressors failing roughly 17× more often than a comparable benchmark unit on the same platform, and trace the cause to a wrong-specification filter element fitted into a housing built for now-discontinued OEM hardware. I am now driving the closeout: OEM and alternative-supplier engagement on a validated fix, packaged into an eleven-action plan with named owners and dates.

Alongside that, I review equipment and system changes, specifications and deviations; contribute to risk screening, HAZID and detailed risk assessments; witness machinery Factory Acceptance Tests; and support commissioning and start-up. I strengthened Oso's Pre-Startup Safety Review process to capture off-skid activities it had missed, and now tailor it to each bundle change-out. I was also lead mechanical and reliability engineer for the restart of the Imo River Associated Gas Gathering plant after roughly two years of prolonged downtime, and I build the digital side of the work too: IIoT condition monitoring that cut operating cost by around ten per cent.

My offshore experience to date is on fixed platforms rather than FPSOs. The machinery scope transfers directly: gas turbine driven compression, turbo-expanders, API 610 and cryogenic pumps, lube-oil and seal systems, and Honeywell CCC anti-surge and turbomachinery control, and I would come up the curve on the floating production context quickly, particularly given the shared engineering heritage of the assets.

I hold a B.Eng in Mechanical Engineering and the NYSC discharge certificate, am an ASQ Certified Reliability Engineer and a Category I vibration analyst, and carry a valid BOSIET with CA-EBS and an Offshore Safety Permit. I am based in Nigeria and available at one month's notice.

Thank you for considering my application. I would welcome the opportunity to discuss how my experience can support machinery reliability and availability across the Nigeria Deepwater FPSOs.

Yours sincerely,

Chukwuemeka Osmund Azodo